TOWER of HANOI

▶ C++ODE

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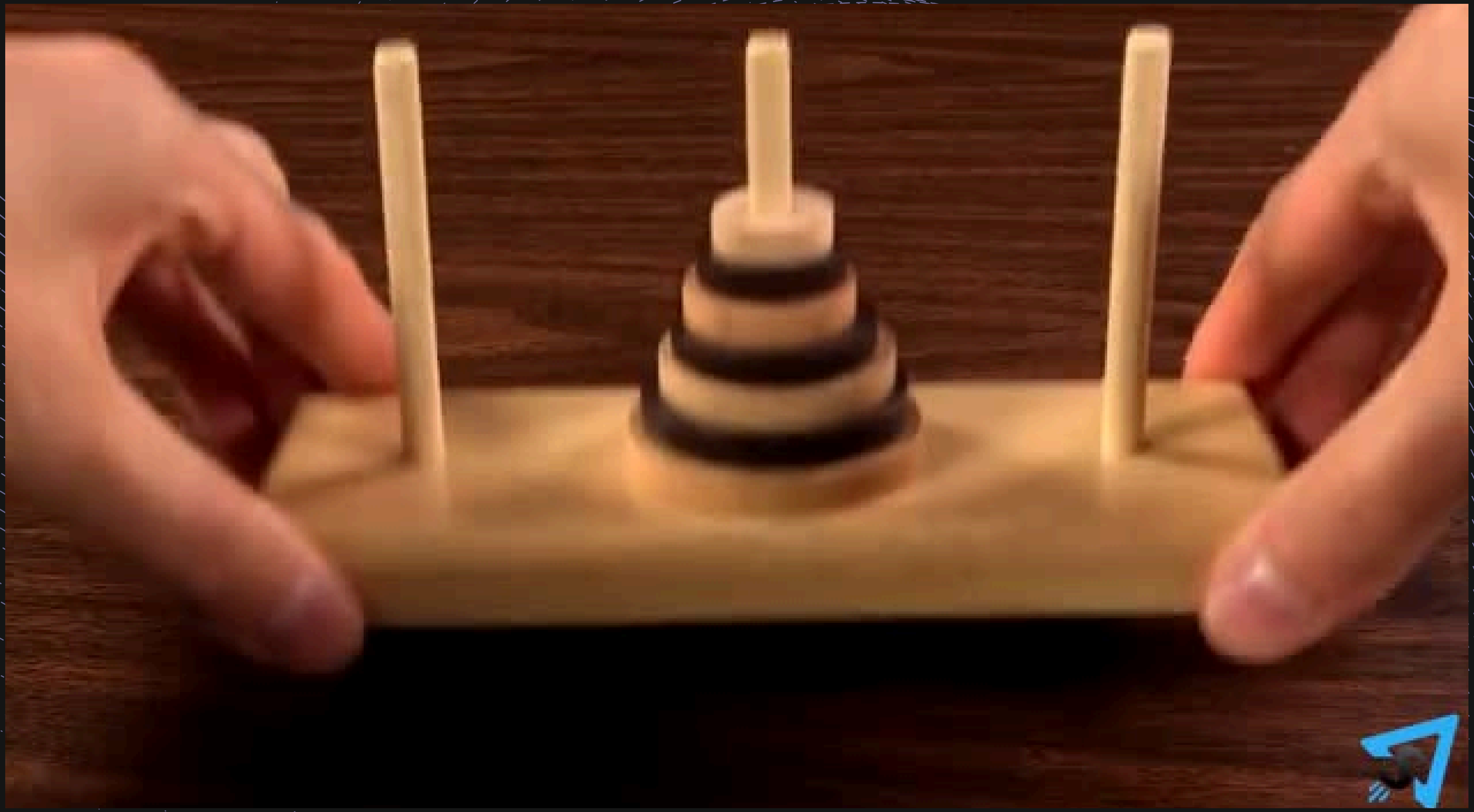
Game concept :



Goal: Console-based code solution to the Tower of Hanoi (3 pegs, 3 disks).

STEP 1

```cpp
#include <iostream>
#include <string>
using namespace std;

int findIdx(int arr[5], int v) {
    for (int i=0; i<5; i++) {
        if (arr[i]==v)
            return i;
    }
    return -1;
}
```

Purpose:

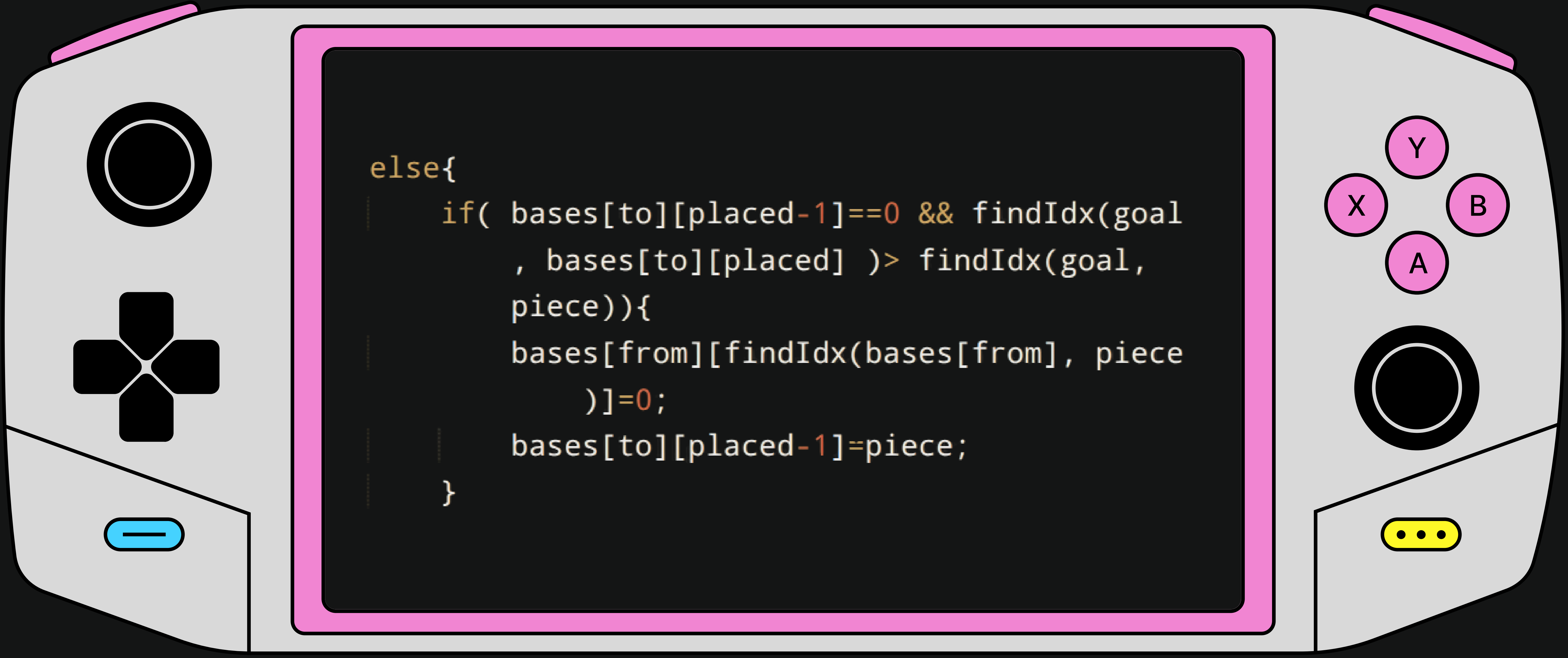
- Imports libraries.
- findIdx searches a fixed-size array and returns index of a value.

▶ STEP 2

```
void shift(int goal[5], int bases[3][5], int move,
    int from, int to ){
    int piece = goal[move-1];
    int placed = 0;
    while(placed<3 && bases[to][placed] ==0){
        placed++;
    }
    if(bases[to][placed-1]==0 && bases[to][placed]
        ==4){
        bases[from][findIdx(bases[from], piece)]=0
            ;
        bases[to][placed-1]=piece;
    }
```

Purpose:

- **shift** attempts to move a disk with rule validation.



else{
if(bases[to][placed-1]==0 && findIdx(goal
, bases[to][placed])> findIdx(goal,
piece)){
bases[from][findIdx(bases[from], piece
)]=0;
bases[to][placed-1]=piece;
}
Y
X
B
A

STEP 3

```cpp
bool check(int x[5], int y[5]){
    for (int i=0;i<4;i++)
        if (x[i]!=y[i])
            return false;
    return true;
}
void display(string basic[5], int base[5]){
    for (int i =0; i<3; i++){
        int ii=base[i];
        cout<<string(7-ii*2,' ')<<basic[ii]<<endl;
    }
    cout<<basic[base[3]]<<endl;
    cout<<string(7,' ')<<char(base[4]+ 97)<<endl;
    cout<<endl;
}
```

Purpose:

- check compares two base states.
- display prints visual representation of one peg.

▶ STEP 4

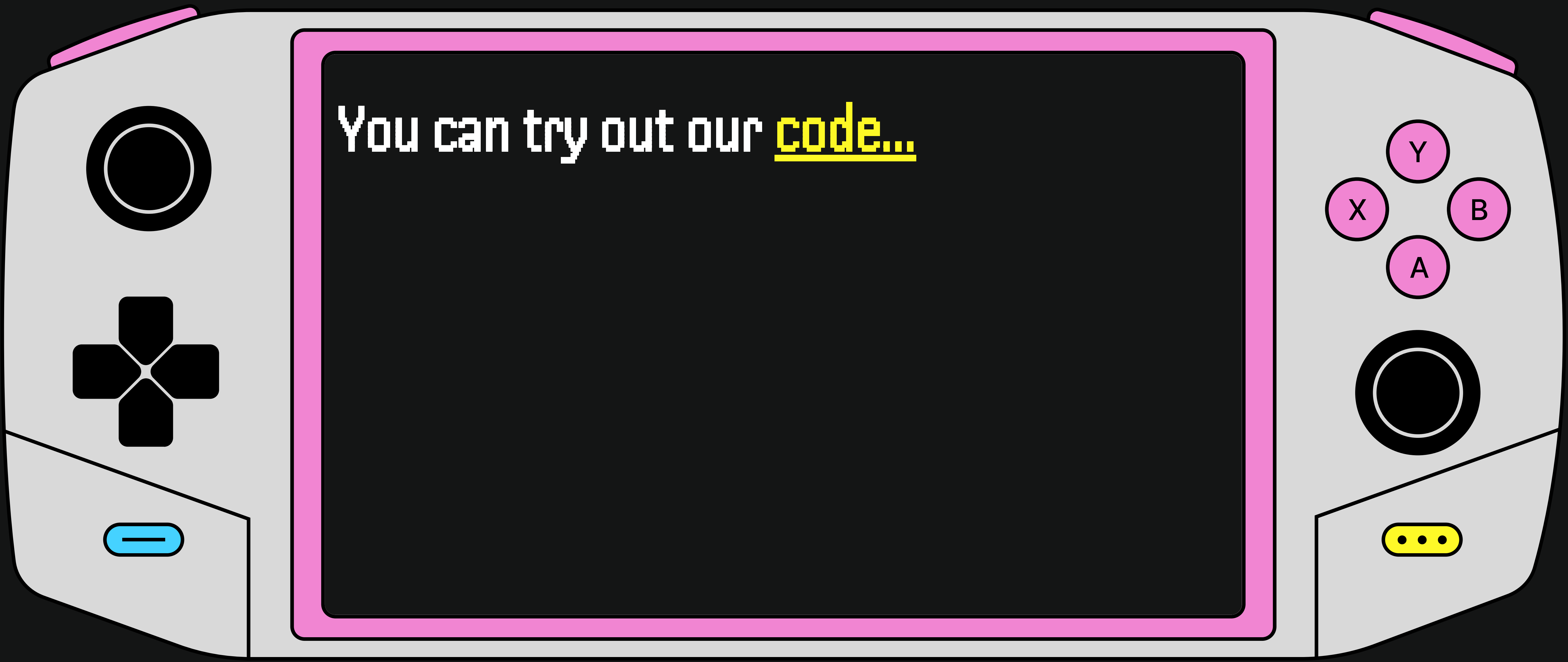
```cpp
int main() {
    string big = "[[[[[[[3]]]]]]]";
    string midium = "[[[[2]]]]";
    string small = "[[1]]";
    string all[5] = {"|", small, midium, big, "[
        /////////////]"};
    int base_a[5] = {1,2,3,4,0};
    int base_b[5] = {0,0,0,4,1};
    int base_c[5] = {0,0,0,4,2};
    int bb[3][5] = {{1,2,3,4,0}, {0,0,0,4,1},{0,0,0
        ,4,2}};
    int goal[5] = {1,2,3,4,0};
    cout<<"Minimum Moves: 7"<<endl<<"Go!"<<endl;
    for (int i = 0; i < 3; i++){
        display(all, bb[i]);
    }
```

Purpose:

- the main fuction handles game loop and user input.

```
int moves =0;
while (!check(bb[2], goal)) {
    cout<<"Your_move!"<<endl;
    int pidx;
    char f,t;
    cout<<"Piece index: ";
    cin>>pidx;
    cout<<"From(a/b/c): ";
    cin>>f;
    int fidx = f-'a';
    cout<<"To(a/b/c): ";
    cin>>t;
    int tidx = t-'a';
    int old_bb[3][5];
    for(int i=0; i<3; i++)
        for(int j=0; j<5; j++)
            old_bb[i][j] = bb[i][j];
```

```
    shift(goal,bb,pidx,fidx,tidx);
    bool valid=false,
    for(int i=0; i<3; i++){
        if(!check(bb[i], old_bb[i])){
            valid=true;
            break;
        }
    }
    if(!valid){
        cout<<"Invalid Move!Try Again"<<endl;
    }
    else{
        for(int n=0; n<3; n++){
            display(all, bb[n]);
        }
        moves++;
        cout<<"move count: "<<moves<<endl;
    cout<<"solved!"<<endl;
    return 0;
```


You can try out our code...
Y
X
B
A


Thank you
FOR YOUR ATTENTION
Y
X
B
A